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[AISEI26] Your paper #1571237682 ('Biometric Identification Through Auditory EEG Signatures')

1 message

Ali Matar <amatar@jadara.edu.jo>

Thu, Mar 12, 2026 at 5:16 AM

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Cc: Ali Matar <amatar@jadara.edu.jo>

Dear Mr. Ly Vu:

Congratulations - your paper #1571237682 ('Biometric Identification Through Auditory EEG Signatures') for AISEI26 has been **accepted** and will be presented in the session titled ____.

The reviews are below or can be found at [1571237682](#).

Blind Review 1**Strengths/Weakness: What are the major reasons to accept/reject the paper? [Be brief.]**

Excellent research paper

Contribution/s & Detailed comments: What are the major issues addressed in the paper? Do you consider them important? Comment on the degree of novelty, creativity and technical depth in the paper. Please provide detailed comments that will be helpful to the TPC for assessing the paper, as well as feedback to the authors.

Healthcare Engineering and Biomedical AI

Originality: New or Novel contribution

Accept (8)

Significance of Topic: Relating to knowledge contribution

Accept (8)

Presentation: Clarity and Organisation of Content

Accept (8)

Blind Review 2**Strengths/Weakness: What are the major reasons to accept/reject the paper? [Be brief.]**

Strengths (Reasons to Accept):

Addresses a Critical Real-World Gap: The paper tackles two significant and practical limitations of current biometric systems: vulnerability to noise and the closed-set assumption. This focus on "open-set" scenarios and "noise-resilience" is highly relevant for real-world deployment.

Technically Sound Approach: The proposed two-stage framework (denoising + metric learning) is logical and leverages state-of-the-art deep learning architectures (WaveNet, ResNet, angular margin losses). The combination is a sensible and promising approach to the problem.

Promising Quantitative Results: The reported performance metrics (Precision@1 up to 98%, TAR@1%FAR up to 95%) are impressive and suggest the framework is highly effective. The analysis on the benefit of decoupling the objectives adds valuable insight.

Novel Application Domain: Applying this specific, robust framework to "brain-derived signals" (EEG) for biometrics is a strong and interesting case study, pushing the boundaries of non-traditional biometrics.

Weaknesses (Reasons to Reject):

Lack of Specificity and Validation (from the Abstract): The abstract is a summary, but for a rigorous review, the absence of concrete details is a major concern. There is no mention of the dataset(s) used, the number of subjects, the type of auditory stimuli, or a comparison against existing state-of-the-art (SOTA) methods. The results, while high, are presented in a vacuum.

Potential for Overclaiming: Terms like "systematically addresses" and "robust solution" are strong. Without a comparative analysis against other noise-robust or open-set methods, such claims cannot be properly evaluated.

Incremental Contribution Risk: While the combination of WaveNet denoising and metric learning for EEG biometrics might be novel, each component individually is well-established. The paper's contribution might be seen as an engineering application of existing tools rather than a fundamental scientific breakthrough, depending on the depth of the technical adaptation required.

Contribution/s & Detailed comments: What are the major issues addressed in the paper? Do you consider them important? Comment on the degree of novelty, creativity and technical depth in the paper. Please provide detailed comments that will be helpful to the TPC for assessing the paper, as well as feedback to the authors.

Major Issues Addressed: The paper addresses three major, interconnected issues in biometric authentication:

Noise Robustness: The vulnerability of biometric systems, especially those using physiological signals like EEG, to various types of environmental and physiological noise.

Open-Set Recognition: The practical requirement for a system to not only identify known individuals but also reliably reject imposters or unknown users, moving beyond the simplified closed-set assumption of most academic studies.

Scalability: The need for a system that can gracefully accommodate new users without full retraining, which is implied by the open-set and metric learning approach.

Importance of the Issues: These issues are of paramount importance for the field. The vast majority of biometric research is conducted in controlled environments under closed-set settings. Translating these successes to real-world, uncontrolled conditions is the grand challenge of the field. Therefore, the paper's focus is on a highly significant and impactful set of problems.

Degree of Novelty, Creativity, and Technical Depth:

Novelty: The novelty appears to lie in the specific combination and application of these techniques. Using a WaveNet for denoising raw (or pre-processed) EEG signals for the purpose of biometric authentication is a creative and well-justified choice. Similarly, applying deep metric learning with angular margins to create an embedding space for EEG-based identity is a strong methodological fit. The paper's own contribution of "explicitly decoupling noise mitigation and embedding learning objectives" is a key insight that demonstrates a thoughtful system design.

Creativity: The creativity is evident in framing the biometric problem as a two-stage pipeline that mirrors a "signal enhancement then discrimination" approach, which is intuitive and effective.

Technical Depth: The abstract suggests significant technical depth. Implementing and optimizing a WaveNet with a scale-invariant SNR objective requires expertise. Integrating this with a ResNet backbone and angular margin loss (like ArcFace or CosFace) for metric learning is non-trivial. The ablation study mentioned ("highlights the importance of explicitly decoupling") points to a rigorous experimental methodology. However, the true depth can only be assessed by reviewing the full paper, including architecture diagrams, training details, and a thorough analysis of failure cases.

Detailed Feedback for Authors (to be included in the review):

The concept and results are compelling. To strengthen the paper, the full manuscript must include a rigorous experimental section. Please specify the EEG dataset(s) used, including size, number of subjects, and the nature of the auditory stimuli.

A direct comparison with several strong baseline methods is crucial. This should include traditional EEG biometric approaches as well as other deep learning-based methods, both for closed-set and open-set scenarios.

Provide more detail on the open-set evaluation protocol. How were unknown users simulated? What were the specific FAR/FRR thresholds?

Elaborate on the "adaptive signal enhancement." How does the WaveNet module adapt to different noise types?

Discuss the computational cost and latency of the two-stage framework, as this is critical for real-time authentication applications.

Originality: New or Novel contribution

Accept (8)

Significance of Topic: Relating to knowledge contribution

Strong Accept (10)

Presentation: Clarity and Organisation of Content

Weak Accept (6)

3/12/26, 2:04 PM

Gmail - [AISEI26] Your paper #1571237682 ('Biometric Identification Through Auditory EEG Signatures')

Regards,
The conference chairs